

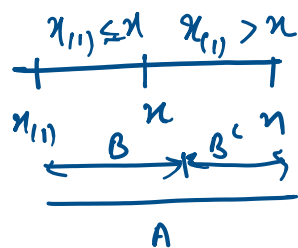
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Joint CDF of $X_{(1)}$ and $X_{(n)}$ order statistics

$$F_{X_{(1)}, X_{(n)}}(x, y) = P(X_{(1)} \leq x, X_{(n)} \leq y) = P(A \cap B) \\ = P(X_{(n)} \leq y) - P(X_{(1)} > x, X_{(n)} \leq y)$$

$$B = \{X_{(n)} \leq y\}, A = \{X_{(1)} \leq x\}$$

$$B^c = \{X_{(n)} > y\}$$



When $x > y$, then $P(X_{(1)} > x, X_{(n)} \leq y) = 0$

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ | \quad | \\ y \quad x \end{array} \Rightarrow F_{X_{(1)}, X_{(n)}}(x, y) = P(X_{(n)} \leq y) = [F(y)]^n$$

$$A = (A \cap B) \cup (A \cap B^c) \\ P(A) = P(A \cap B) + P(A \cap B^c) \\ P(A \cap B) = P(A) - P(A \cap B^c)$$

When $x < y$, $P(X_{(1)} > x, X_{(n)} \leq y) = P(x < X_1 \leq y, \dots, x < X_n \leq y)$

$$\begin{aligned} \min\{X_1, \dots, X_n\} > x, \max\{X_1, \dots, X_n\} \leq y &= (P(x < X_1 \leq y))^n \\ x_1 > x, x_2 > x, \dots, x_n > x, x_1 \leq y, \dots, x_n \leq y &= (P(X_1 \leq y) - P(X_1 \leq x))^n \\ &= (F(y) - F(x))^n \end{aligned}$$

$$\Rightarrow F_{X_{(1)}, X_{(n)}}(x, y) = (F(y))^n - (F(y) - F(x))^n$$

Joint PDF (for $x < y$)

$$f_{X_{(1)}, X_{(n)}}(x, y) = \frac{\partial}{\partial x} \frac{\partial}{\partial y} \{F_{X_{(1)}, X_{(n)}}(x, y)\}$$

$$= \frac{\partial}{\partial x} \{n F(y)^{n-1} f(y) - n(F(y) - F(x))^{n-1} f(y)\}$$

$$= 0 + n(n-1) [F(y) - F(x)]^{n-2} f(x) f(y)$$

$$f_{X_{(1)}, X_{(n)}}(x, y) = n(n-1) [F(y) - F(x)]^{n-2} f(x) f(y), \quad x < y$$

(*)

Joint pdf of $X_{(i)}$ and $X_{(j)}$ ($i \leq j$)

$$f(x, y) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} [F(x)]^{i-1} [F(y) - F(x)]^{j-i-1} [1 - F(y)]^{n-j}$$

— (**) $f(x)f(y)$

Special case :- $i=1, j=n$

Joint PDF of all order statistic :-

$$f_{X_{(1)}, \dots, X_{(n)}}(x_1, x_2, \dots, x_n) = n! f(x_1) f(x_2) \dots f(x_n), \quad -\infty < x_1 < x_2 < \dots < x_n < \infty$$

Q1:- Let $X_{(1)} \leq X_{(2)} \leq X_{(3)} \leq X_{(4)}$ be order statistic of a random sample of size 4 with pdf $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{else} \end{cases}$
Find joint pdf of $X_{(3)}$ and $X_{(4)}$.

Ans:- $f_{X_{(3)}, X_{(4)}}(x_3, x_4) = 20 x_3^5 x_4, \quad 0 < x_3 \leq x_4 < 1$

Q2:- Consider 15 RV, $X_i \sim U[0, 1]$, $i=1, 2, \dots, 15$. Find joint pdf of $X_{(1)}$ and $X_{(15)}$.

Q3:- Let $X_i \sim U[0, 1]$ be i.i.d, find the pdf of j th order statistics and find its mean and variance.

Ans:- $f_{X_{(j)}}(x) = \frac{1}{B(j, n-j+1)} x^{j-1} (1-x)^{n-j}$

$X_{(j)} \sim \text{Beta}(j, n-j+1)$

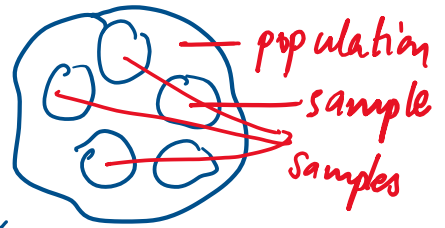
$$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

$$= \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Sampling Distributions:-

① Population:- A collection of measurements
In other words, a group that we are interested in studying.

② Sample:- A subset of population



③ Random sampling:- Let $X_1, X_2, \dots, X_n$ be n iid RVs each having the same prob. distⁿ $f(x)$. Then we say $(X_1, X_2, \dots, X_n)$ is a random sample from popⁿ with distⁿ $f(x)$ and joint distⁿ of $X_1, X_2, \dots, X_n$ is defined as $f(x_1, x_2, \dots, x_n) = f(x_1) \cdot f(x_2) \dots f(x_n)$

④ Statistic:- A fⁿ of random sample $T = T(X_1, X_2, \dots, X_n)$ is called a statistic.

Ex:- (a) sample mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

(b) sample variance: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

(c) sample range: $X_{(n)} - X_{(1)}$

Note:-



⊕ sample

parameters: mean(μ), variance(σ^2)

statistic:- $\bar{X}, S^2$
↓
used to estimate the parameter.

⑤ Sampling Distribution:- The prob. distⁿ of a statistic is called a sampling distⁿ

Remark:- Some particular statistic like sample mean, sample variance, etc play vital role. So, we will study the distⁿ of these statistic.

Central Limit Theorem:- Let $X_1, X_2, \dots, X_n$ be a seq of iid RVs with mean μ and variance σ^2 ($< \infty$). Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, then limiting distⁿ of $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$ as $n \rightarrow \infty$

Gamma Distribution:- A RV is said to have a gamma distⁿ with parameters (α, λ) , $\alpha > 0$, $\lambda > 0$ if its density f^n is given by

$$G(\alpha, \lambda) = f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

$$\Gamma(\alpha) = \int_0^{\infty} e^{-y} y^{\alpha-1} dy, \quad y = \lambda x$$

Check:- For $\alpha = 1$, $G(1, \lambda) \sim ?$ $G(1, \lambda) \sim \text{Exp}(\lambda)$

$$\textcircled{1} \Gamma(1) = 1, \quad \textcircled{2} \Gamma(n+1) = n\Gamma(n), \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$
$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

Other sampling distⁿ:-

(1) Chi-square distⁿ (χ^2):- A cont. RV W is said to have a chi-sqⁿ distⁿ with " n " degree of freedom if it has the following pdf

$$\boxed{X \sim \chi_n^2}$$

$$f(w) = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-w/2} w^{\frac{n}{2}-1}, w > 0, n > 0$$

Note i- $\alpha = \frac{n}{2}$, $\lambda = \frac{1}{2}$ in $G(\alpha, \lambda) \sim \chi_n^2$

Thm i- If $W_1 \sim \chi_{n_1}^2$, $W_2 \sim \chi_{n_2}^2$... $W_k \sim \chi_{n_k}^2$ where $W_1, W_2, \dots$

... W_k are independent, then

$$W_1 + W_2 + \dots + W_k \sim \chi_{n_1 + n_2 + \dots + n_k}^2$$

Ex i- $W_1 \sim \chi_3^2$, $W_2 \sim \chi_4^2$, $W_3 \sim \chi_2^2$,

$$W_1 + W_2 + W_3 \sim \chi_9^2$$